BLOCK DIFFUSION IS MORE COMPUTE-EFFICIENT THAN STANDARD DIFFUSION: ISOFLOP EVIDENCE AND CURRICULUM LEARNING ON TINYSHAKESPEARE

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ABSTRACT

Discrete diffusion language models are promising but remain at least an order of magnitude less FLOP-efficient than autoregressive (AR) models. We present controlled IsoFLOP experiments on character-level TinyShakespeare comparing four diffusion objectives in both standard and block formulations. At a budget of 6×10^{14} FLOPs, block diffusion (block length 4) outperforms standard diffusion by 0.05–0.16 nats, with Block MDLM achieving 1.824 vs. 1.981 for standard MDLM. A curriculum that warm-starts with AR before switching to block diffusion yields an additional 0.04–0.09 nat improvement, with a sweet spot at $p_{\text{AR}} = 0.3$.

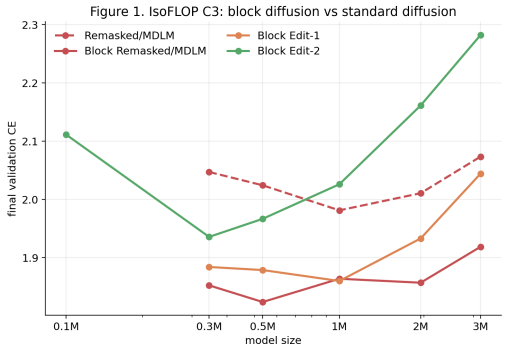
1 INTRODUCTION

Diffusion language models (dLMs) offer flexible, non-left-to-right generation but are substantially less compute-efficient to pretrain than autoregressive (AR) models (Sahoo et al., 2024; Shi et al., 2024). Masked diffusion (MDLM) is roughly $14\times$ less efficient than AR; uniform-noise diffusion is roughly $32\times$ less efficient (Nie et al., 2025a). Block diffusion (Arriola et al., 2025) interpolates between AR and diffusion by splitting the sequence into blocks denoised left-to-right with cross-block causal attention. Meanwhile, recent work converts pretrained AR checkpoints into diffusion models (various, 2025b;a; Ye et al., 2025; Nie et al., 2025b), but the optimal *curriculum* remains poorly understood. We present two sets of experiments on TinyShakespeare (Karpathy, 2015): (1) IsoFLOP comparisons of block vs. standard diffusion across four objectives, and (2) a curriculum study warm-starting block diffusion with AR pretraining.

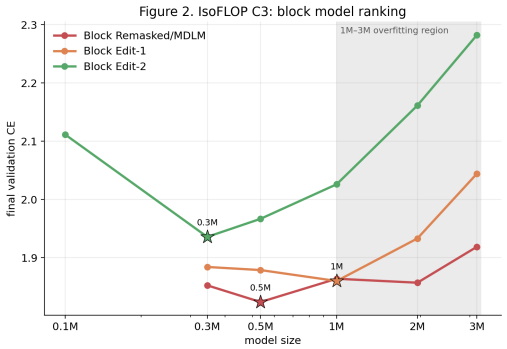
2 SETUP

All nine model variants share a transformer backbone with RoPE, QK-norm, ReGLU MLPs, and RMSNorm (n_{head}=4), spanning six sizes from 0.1M (d=64, 2 layers) to 3M (d=256, 4 layers). We study four objectives: **MDLM** (progressive unmasking), **Remasked** (re-masks low-confidence tokens at generation; identical training loss to MDLM), **Edit-1** (trains on masked + 15%-corrupted visible tokens), and **Edit-2** (two-pass correction, $2\times$ FLOP cost). Each has a block counterpart (block length 4) using dual-stream attention during training.

We fix a single IsoFLOP budget $C_3 = 6 \times 10^{14}$ and compute training steps as $C_3/(C \cdot N \cdot 32768)$, where C is a FLOP multiplier (C=6 for standard, 12 for block, 24 for block Edit-2) and 32768 tokens are processed per step. Learning rates were determined by a prior two-stage sweep ($10\times$ then $3\times$ zoom); all runs use linear warmup with cosine decay.

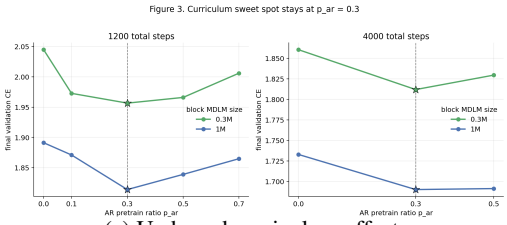


(a) Block vs. standard diffusion at C_3.

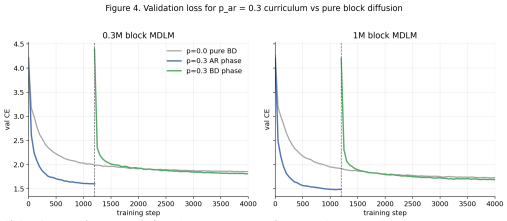


(b) Block model ranking; shaded = overfitting.

Figure 1: IsoFLOP curves at $C_3 = 6 \times 10^{14}$ FLOPs on TinyShakespeare.



(a) U-shaped curriculum effect.



(b) AR phase (blue) $\rightarrow$ BD phase (green) vs. pure BD (gray).

Figure 2: Curriculum learning: AR warm-start for block MDLM.

3 Results

Block diffusion is more FLOP-efficient. Figure 1(a) shows that at equal compute, all block models outperform their standard counterparts. Block MDLM at 0.5M achieves validation CE 1.824, vs. 1.981 for standard MDLM at 1M—a 0.16-nat gap. This advantage holds despite the $2\times$ per-step cost of block training, which is offset by the inductive bias of left-to-right block generation.

Block models prefer smaller sizes at fixed compute. Block MDLM's IsoFLOP optimum is at 0.5M versus 1M for standard MDLM (Figure 1(b)). Beyond 0.5M, block models get too few steps under the fixed budget and overfit. The ranking is: Block MDLM/Remasked (1.824) $>$ Block Edit-1 (1.860) $>$ Block Edit-2 (1.936).

AR pretraining helps, with sweet spot at $p_{\text{AR}} = 0.3$. We train AR for a fraction p_{AR} of total steps, then continue as block MDLM for the rest (the weight transfer is seamless since both share the same backbone). Table 1 shows that $p_{\text{AR}} = 0.3$ consistently gives the lowest validation loss: -0.088 nats at 0.3M/1200 steps and -0.043 nats at 1M/4000 steps. The effect is U-shaped (Figure 2(a)): too little AR undertrains the initialization; too much starves the diffusion phase. At 3M the effect vanishes, likely because the model converges within budget regardless. Figure 2(b) shows that the AR phase provides a strong initialization and the subsequent block diffusion loss stays below the pure-BD baseline throughout.

4 Discussion

Our experiments yield three actionable findings: (1) block diffusion provides a meaningful FLOP-efficiency advantage over standard diffusion, (2) the IsoFLOP-optimal size shifts smaller for block models due to their higher per-step cost, and (3) a 30% AR warm-start robustly improves block diffusion. Key next steps include validating at OpenWebText scale (10M–300M parameters) with multiple IsoFLOP budgets for Chinchilla-style scaling curves (Hoffmann et al., 2022), sweeping the noise-schedule clipping parameters $t_{\min}$ and $t_{\max}$ (fixed at 0.45 and 0.95 here), exploring AR$\rightarrow$block-uniform curricula motivated by the downstream advantages of uniform-noise diffusion (Nie et al., 2025a), and adopting a warmup-stable-decay LR schedule for the curriculum transition.

Table 1: Validation CE for AR→Block MDLM curriculum. Bold = best per row.

	1200 total steps					
Size	p=0	p=0.1	p=0.3	p=0.5	p=0.7	Δ
0.3M	2.045	1.973	**1.957**	1.966	2.006	−0.088
1M	1.891	1.871	**1.814**	1.839	1.865	−0.077
	4000 total steps					
Size	p=0	p=0.3	p=0.5			Δ
0.3M	1.860	**1.812**	1.830			−0.049
1M	1.733	**1.690**	1.692			−0.043

ACKNOWLEDGMENTS

Experiments were run on a single NVIDIA A100 GPU.

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